

Intended Learning Objectives

In this lab you will:

- Learn how to generate and use g_m/I_D design curves.
- Learn how to design a two-stage Miller OTA achieving given specifications.
- Learn how to simulate the open-loop characteristics of the two-stage Miller OTA.
- Learn how to simulate the closed-loop characteristics of the two-stage Miller OTA.

PART 1: g_m/I_D Design Charts

Download ADT:

- Go to <https://adt.master-micro.com>
- Register using your university or corporate email address. If you are a student or fresh grad, select academia as your organization type. If you don't have a university or corporate email address then register as unemployed and include your LinkedIn profile URL, but your account may take some time to get reviewed and approved.
- Read ADT readme file. Visit ADT website again and generate a free personal license.

Using ADT Device Xplore, plot the following design charts vs g_m/I_D for both PMOS and NMOS. Set $V_{DS} = V_{DD}/3$ and $L = 0.18\mu, 0.25\mu, 0.5\mu, 2\mu$

1. g_m/g_{ds}
2. I_D/W
3. g_m/C_{gg} (use advanced Y expression)
4. V_{GS}

PART 2: OTA Design

Use g_m/I_D methodology to design a differential input, Single Ended output **two-stage Miller-compensated OTA**. The OTA is to be used as a buffer (unity gain feedback configuration) to probe sensitive internal signals in a complex mixed-signal design. The OTA should achieve the specs below.

Parameter	Value
Supply voltage	1.8 V
Static gain error	$\leq 0.05\%$
CMRR @ DC	≥ 74 dB
Phase margin (avoid pole-zero doublets)	$\geq 70^\circ$
OTA current consumption	$\leq 60\ \mu\text{A}$
CMIR - low	≤ 0.2 V
CMIR - high	≥ 1 V
Output Swing	0.2 V – 1.6 V
Load	5 pF
Buffer closed loop rise time (10% to 90%)	≤ 70 ns
Slew Rate	5 V/ μs

Use an ideal external $10\ \mu\text{A}$ DC current source in your test bench (not included in the OTA current consumption spec), but design your own bias circuit (current mirrors). Create a schematic and an appropriate symbol for the OTA.

► **Suggested Design Procedure (you may create your own!):**

1. Use a single $10\ \mu\text{A}$ DC current source and a single DC voltage source in your test bench. Design your own current mirrors and bias circuitry.
2. A reasonable starting point for C_c is $0.5C_L$. You may refine this choice by doing sweeps in simulation.
3. Calculate the unity gain frequency (UGF) from the rise time requirement ($t_{\text{rise}} = 2.2\tau$). Hence, calculate $g_{m1,2}$.
4. From the SR requirement, calculate the current required in the first stage (I_{B1}): $\text{SR} = \frac{I_{B1}}{C_c}$. Given the total current budget, calculate the current of the second stage.
5. Calculate g_m/I_D of the first stage.
6. Show that the closed-loop gain for a buffer is $A_{\text{VCL}} \approx 1 - \frac{1}{A_{\text{VOL}}}$ where A_{VOL} is the open-loop gain. Given A_{VCL} gain error spec $\left(\%\text{error} = \left| \frac{\text{actual} - \text{ideal}}{\text{ideal}} \right| \times 100 \right)$, calculate the required DC gain in dB.
7. Assign larger gain for the first stage (why?). Do not split the gain equally between the two stages. You may assume the first stage gain is twice that of the second stage (6dB difference).
8. Given the 1st stage gain, calculate L (channel length) of the 1st stage input. You may assume input and load have the same g_{ds} .
9. Given V_A of the first stage current mirror load, select L . Note that V_A slightly decreases with g_m/I_D , which is not known yet. To get an estimate for L , you may ignore this dependence and assume a relatively large g_m/I_D for the load at this point (e.g., $g_m/I_D = 15$).

(This step is needed if you are using design charts with ' L ' as a parameter. You can skip this step if you are using the ADT Sizing Assistant and use the g_{ds} directly as your condition.)

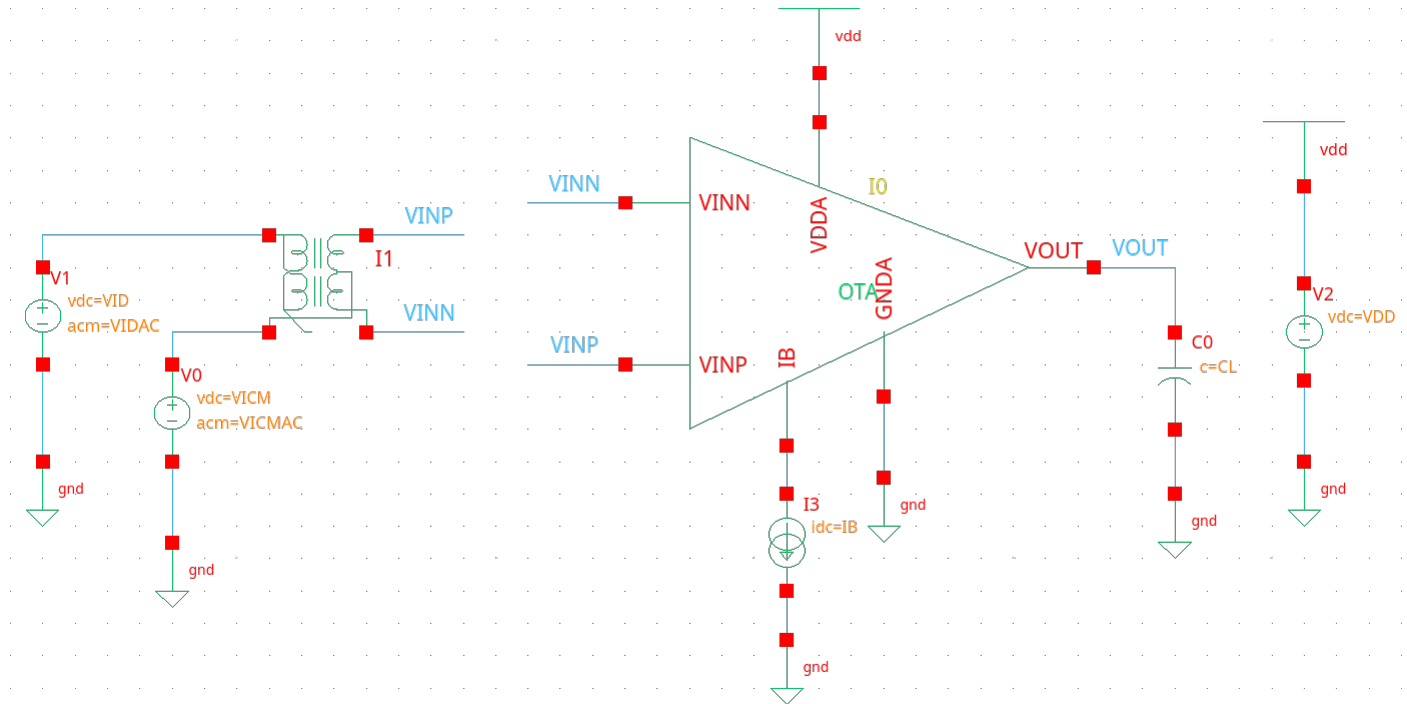
10. Given the PM spec, calculate g_m/I_D of the second stage input transistor (Hint: assume $\omega_{p2} = 4\omega_u$).
11. Given the CMIR-high and Swing-high specs, calculate $\max v_{dsat}$ for tail current source and output load. Take the lower value and assume $V^* = v_{dsat}$. Note that always $V^* > v_{dsat}$; thus, this assumption already adds some margin to make sure they are driven a little more into saturation. Now you have g_m/I_D of these two transistors. Note that these two transistors are identical (they form a current mirror; thus, they have same L and same g_m/I_D) except for the current (and width).
12. Use the CMRR spec to find g_{ds} of the tail current source (note that the second stage does not affect the CMRR). However, to complete this step you need g_m of the current mirror load. This is not known yet, because we want V_{GS} of 1st stage load = V_{GS} of 2nd stage input. To break this deadlock, assume a relatively low g_m/I_D (e.g., $g_m/I_D = 10$) for first stage current mirror load. Thus, get g_{ds} of tail current source.

(For current mirror load, we first assumed $g_m/I_D = 15$ then we assumed $g_m/I_D = 10$. Is this contradicting? No. For the DC gain, we assumed a bit high $g_m/I_D = 15$ because if it is eventually a lower value (< 15) V_A will be higher ($g_m/I_D \uparrow \Rightarrow V_A \downarrow$) and the gain will be even better than the spec (positive error). Next, for the CMRR (A_{vcm}) we assumed a bit low $g_m/I_D = 10$ because if it eventually higher (> 10) A_{vcm} will be smaller and the CMRR will be even better than the spec (positive error). If eventually $g_m/I_D < 10$ or > 15 then we will have to reiterate.)
13. Tail current source and 2nd stage load must have the same L (they form a current mirror). Thus, get g_{ds} of the 2nd stage load (note that both g_m and g_{ds} are proportional to I_D).
14. Given the 2nd stage gain, calculate g_{ds} and L of the 2nd stage input transistor. This transistor is now fully specified; thus, calculate its V_{GS} .
15. Note that you need to avoid systematic offset. Use V_{GS} charts to guarantee that 1st stage current mirror load and 2nd stage input transistor have the same V_{GS} . **Use this condition to determine the g_m/I_D of the current mirror load in the first stage.** Check that the calculated g_m/I_D is larger than the one you assumed before (to guarantee that the CMRR is satisfied). Otherwise, re-iterate with the new g_m/I_D value (if the difference is large). If this step is not done properly, you will find that V_{OUT} goes towards V_{DD} or GND and one of the output transistors is out of saturation. **In order to make sure that the systematic offset is canceled, you can sweep the width of the current mirror load with fine step till V_{OUT} is around $V_{DD}/2$.**
16. Verify that your g_m/I_D choices do not violate the CMIR and the peak-to-peak output swing.
17. Choose R_Z to place the zero at infinity (some designers may move the LHP zero to the vicinity of the non-dominant pole to improve the PM). Do **NOT** place the LHP zero at a frequency less than ω_{p2} .

Report the following:

1. Detailed design procedure and hand analysis. Justify why you used NMOS or PMOS input pair for each stage.
2. A table showing W , L , g_m , I_D , g_m/I_D , v_{dsat} , $V_{ov} = V_{GS} - V_{TH}$, and $V^* = 2I_D / g_m$ of all transistors (as calculated from g_m/I_D curves).

PART 3: Open-Loop OTA Simulation



Create a testbench similar to the one shown above (the shown schematic is from the 5T OTA lab). Note that IDC connection in the test bench (sinking or sourcing) may be different from the one shown above depending on the type of your input pair (PMOS/NMOS).

NOTE: The open-loop simulation will NOT work UNLESS there is ZERO offset voltage.

Report the following:

1. Schematic of the OTA and bias circuit with DC node voltages clearly annotated.

- Use $V_{ICM} = V_{DD}/3$.
- Is the current (and gm) in the input pair exactly equal?
- What is DC voltage at the output of the first stage? Why?
- What is DC voltage at the output of the second stage? Why?

2. Diff small signal ccs:

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set $V_{IDAC} = 1$ and $V_{ICMAC} = 0$.
- Use $V_{ICM} = V_{DD}/3$.
 - **Cadence Hint:** Use Cadence calculator expressions to calculate circuit parameters (A_o , A_o in dB, BW, GBW, UGF). You may use Cadence calculator to create other useful expressions.

Name	Type	Details	Plot
A0	expr	$\text{ymax}(\text{mag}(\text{VF}/\text{VOUT}))$	☒
A0_dB	expr	$\text{dB20}(\text{ymax}(\text{mag}(\text{VF}/\text{VOUT})))$	☒
BW	expr	$\text{bandwidth}(\text{VF}/\text{VOUT}) \cdot 3 \cdot \text{"low"}$	☒
fu	expr	$\text{unityGainFreq}(\text{VF}/\text{VOUT})$	☒
	expr	$\text{dB20}(\text{mag}(\text{VF}/\text{VOUT}))$	☒
	expr	$\text{phase}(\text{VF}/\text{VOUT})$	☒
	expr	$\text{phaseMargin}(\text{VF}/\text{VOUT})$	☒
GBW	expr	$(\text{A0} \cdot \text{BW})$	☒

- Plot diff gain (in dB) vs frequency.
- Compare simulation results with hand calculations in a table.

3. CM small signal ccs:

- Use AC analysis (1Hz:10GHz, logarithmic, 10 points/decade).
- Set $V_{\text{ICMAC}} = 1$ and $V_{\text{IDAC}} = 0$.
- Use $V_{\text{ICM}} = V_{\text{DD}}/3$.
- Plot CM gain in dB vs frequency.
- Compare simulation results with hand calculations in a table.

4. CMRR:

► **Cadence Hint:** In Mentor Pyxis you have to get A_{vd} and A_{vcm} from two independent simulation runs (we cannot run both simultaneously because we have single ended output, thus we cannot differentiate between diff and CM signals at the output). But for Cadence Virtuoso you should use XF analysis (1Hz:10GHz, logarithmic, 10 points/decade) because you need to calculate the transfer function between multiple inputs and a single output.

► **Cadence Hint:** Access XF analysis results from the results browser or from adexl results tab (Right Click -> Direct Plot -> Main Form). You may use this expression in the calculator to plot the CMRR: $\text{dB20}(\text{mag}(\text{getData}("/\text{V2}" ?\text{result} "xf")))) - \text{dB20}(\text{mag}(\text{getData}("/\text{V1}" ?\text{result} "xf"))))$.

- Use $V_{\text{ICM}} = V_{\text{DD}}/3$.
- Plot CMRR in dB vs frequency.
- Compare simulation results with hand calculations in a table.

5. (Optional) Diff large signal ccs:

- Use $V_{\text{ICM}} = V_{\text{DD}}/3$.
- Use DC sweep (**not parametric sweep**) $V_{\text{ID}} = -0.1:1\text{m}:0.1$. You must use a small step because the gain region is very small (steep slope).
- From the plot, what is the value of V_{OUT} at $V_{\text{ID}} = 0$. Compare it with the value you obtained in DC OP.
- Plot V_{OUT} vs V_{ID} .

- Plot the derivative of V_{OUT} vs V_{ID} . Is the peak less than the value of A_{vd} obtained from AC analysis? Why?

6. CM large signal ccs (region vs V_{ICM}):

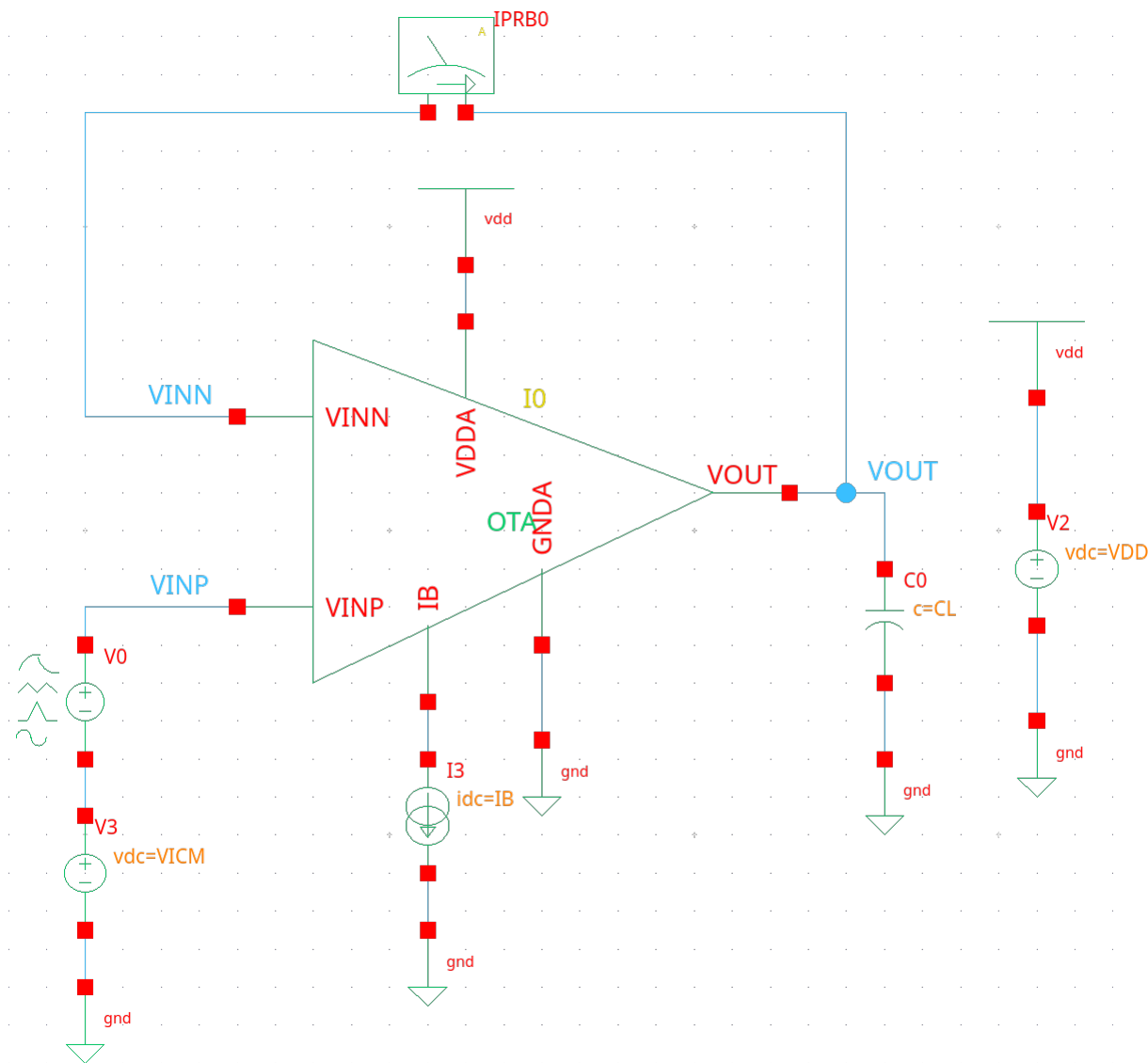
- Use **DC sweep** (not parametric sweep) $V_{ICM} = 0:10m:V_{DD}$.
- Plot “region” OP parameter vs V_{ICM} for the input pair and the tail current source (0 cut-off, 1 triode, 2 sat, 3 subth, and 4 breakdown).
- Plot “region” OP parameter vs V_{ICM} for the input pair and the tail current source.
- Find the CM input range (CMIR). Compare with hand analysis in a table.
- Note that the drawback of this method is that the “region” parameter cannot be experimentally measured in the lab.

7. (Optional) CM large signal ccs (GBW vs V_{ICM}):

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set $V_{IDAC} = 1$ and $V_{ICMAC} = 0$.
- Use parametric sweep (not DC sweep) $V_{ICM} = 0:25m:V_{DD}$.
- Plot GBW vs V_{ICM} . Plot the results overlaid on the results of the next method (region parameter).
 - **Cadence Hint:** Instead of using parametric sweep, a better alternative in Cadence is to use AC sweep but sweep a design variable (V_{ICM}) instead of sweeping the frequency. The frequency is set at 1 Hz (or any other small value) to get the low frequency gain (DC gain).
- Annotate the CM input range. Calculate the input range as the range over which the GBW is within 90% of the max GBW, i.e., 10% reduction in GBW.

(If you are using NMOS input pair, body effect may cause CMIR to extend till V_{DD} (why?). If you are using PMOS input pair, body effect may cause CMIR to extend till GND (why?).)

PART 4: Closed-Loop OTA Simulation



Create a new testbench with the OTA connected in a unity gain buffer feedback configuration (the shown schematic is from the 5T OTA lab). Place a current probe (iprobe) or a zero voltage source in the feedback loop.

Report the following:

1. Schematic of the OTA and the bias circuit with DC OP point clearly annotated in unity gain buffer configuration.
 - Use $V_{ICM} = V_{DD}/3$.
 - Are the DC voltages at the input terminals of the op-amp exactly equal? Why?
 - Is the DC voltage at the output of the first stage exactly equal to the value in the open-loop simulation? Why?
 - Is the current (and g_m) in the input pair exactly equal? Why?
2. Loop gain:
 - Use STB analysis (1Hz:10GHz, logarithmic, 10 points/decade) in unity gain buffer configuration.

- Use V_{ICM} at the middle of the CMIR.
- Plot loop gain in dB and phase vs frequency.
 - **Cadence Hint:** Access STB analysis results from the results browser or from adexl results tab (Right Click in adexl Results tab-> Direct Plot -> Main Form)
- Compare DC gain, f_u , and GBW with those obtained from open-loop simulation. Comment
- Report PM. Compare with hand calculations. Comment.
- Compare simulation results with hand calculations in a table.

3. Slew rate:

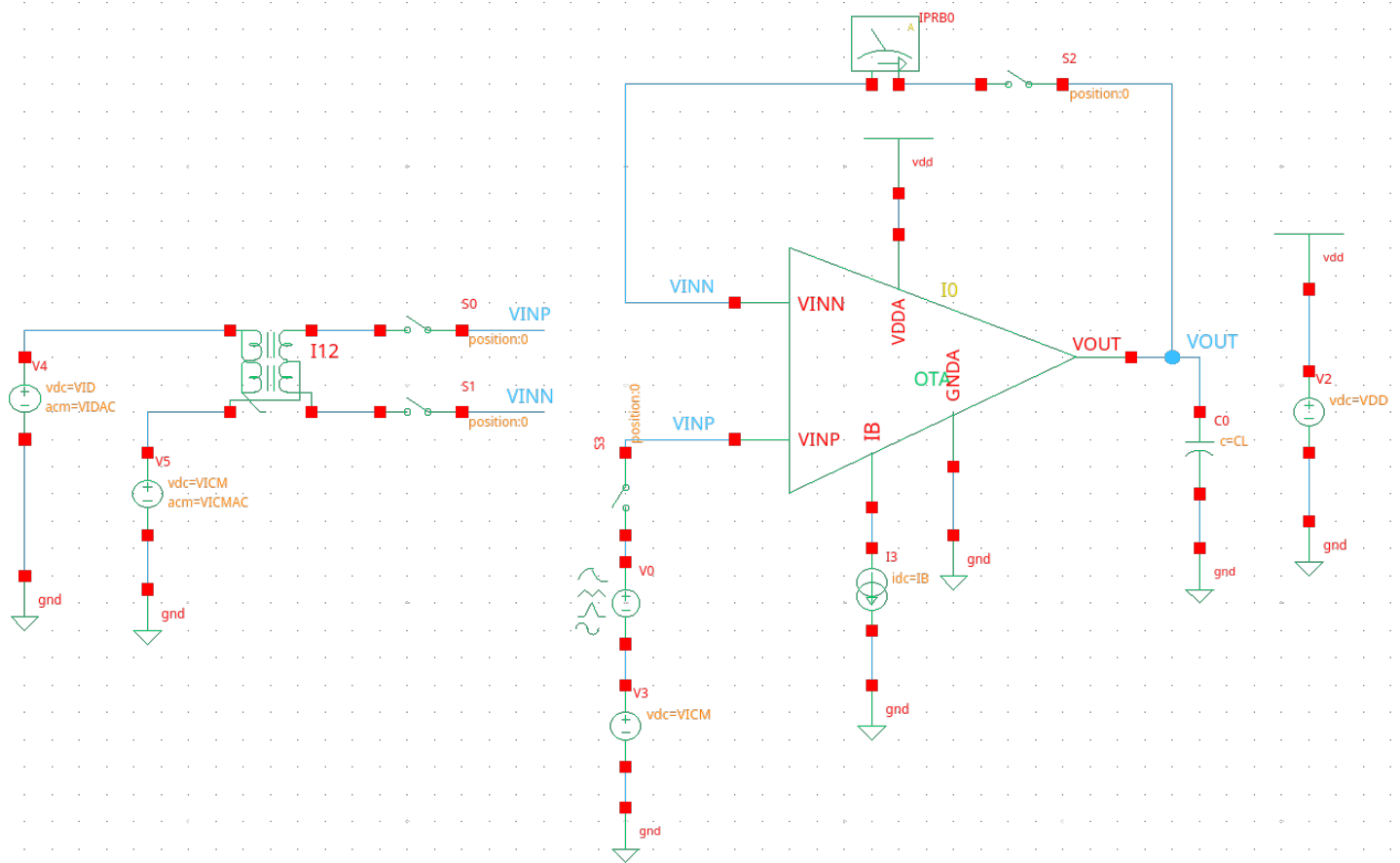
- Apply a step input with the following parameters (delay = $1\mu s$, initial value = CMIR-low + 50mV, final value = CMIR-high – 50mV, rise time = 1ns, period = 1s, width = 1s). Note that we want a single step input, which is why we selected very large period and width for the pulse.
- Run transient analysis (stop = $5\mu s$ and step = 0.1ns).
- Report V_{in} and V_{out} overlaid.
- Report the slew rate.
- Compare simulation results with hand calculations in a table.

4. Settling time:

- Apply a small signal step input with the following parameters (delay = $1\mu s$, initial value = $V_{DD}/3$, final value = $V_{DD}/3 + 5mV$, rise time = 1ns, period = 1s, width = 1s). Note that we want a single step input, which is why we selected very large period and width for the pulse. Note that we apply a small signal pulse (5mV step) to measure the small signal settling time.
- Calculate the output rise time from simulation.
- Compare simulation results with hand calculations in a table.
(The simulation result will be better than expected. Why? (Hint: Using $t_{rise} = 2.2\tau$ is based on first-order model. Is second-order system faster?))
- Do you see any ringing? Why?

PART 5 (optional): DC Closed Loop AC Open-Loop OTA Simulation

Note that there will always be residual offset voltage between the first and second stages. This offset will drive the second stage output to one of the rails; thus, the biasing of the output stage will be disturbed. In order to avoid this problem, the DC OP point must be set by feedback, e.g., by putting the amplifier in unity gain buffer configuration in DC. Use a testbench similar to the one used in the 5T OTA lab as shown below.



Switches (sp1tswitch from analogLib library) are added in order to connect the feedback loop in DC and break it in AC (another option is to use resistors with different AC/DC values). Set S0 and S1 DC closed, and set S2 and S3 AC closed. The DC OP point is set by the unity gain feedback buffer connection, while the AC stimulus is set by the balun.

Lab Summary

- In Part 1 you learned
 - How to generate and use g_m/I_D design curves.
- In Part 2 you learned
 - How to design two-stage Miller OTA meeting desired specifications.
- In Part 3 you learned
 - How to simulate the small-signal differential gain of two-stage Miller OTA in open-loop configuration.
 - How to simulate the small-signal common-mode gain of two-stage Miller OTA in open-loop configuration.
 - How to simulate the large-signal differential characteristics of two-stage Miller OTA in open-loop configuration.
 - How to simulate the large-signal common-mode characteristics of two-stage Miller OTA in open-loop configuration.

- In Part 4 you learned
 - How to simulate the small-signal differential gain of two-stage Miller OTA in closed-loop configuration.
 - How to simulate the small-signal common-mode gain of two-stage Miller OTA in closed-loop configuration.
 - How to simulate the large-signal differential characteristics of two-stage Miller OTA in closed-loop configuration.
 - How to simulate the large-signal common-mode characteristics of two-stage Miller OTA in closed-loop configuration.
- In Part 5 you learned
 - How to use DC closed-loop configuration to simulate an AC open-loop configuration.